

Synopsis

Synopsis – Women Safety Android Application

1. Title

Women Safety Application (Java)

2. Introduction

Safety of women has become a major concern in today's society. This project aims to develop a Women Safety Mobile Application using Java in Android Studio, which helps women in emergency situations and also provides resources for awareness and protection. The app ensures quick access to SOS features, emergency services, safety articles, videos, and experience sharing.

3. Objectives

To provide an SOS alert system that works on button click or shake detection.

To send real-time location and alert messages to registered emergency contacts.

To provide quick dial access to emergency services such as Police, Ambulance, and Fire.

To locate nearby police stations, hospitals, and ambulance services via Google Maps.

To spread awareness through articles and self-defense tutorial videos.

To build a review & experience sharing section, where users can post and view unsafe incidents/areas.

To ensure secure access using Firebase Authentication.

4. Modules / Features

1. Authentication Module

Login & Signup using Firebase Authentication.

2. Home Page Module

Central hub for all safety features.

3. SOS & Shake Detection Module

Detects phone shake or SOS button click.

Sends SMS with live GPS location to emergency contacts.

4. Emergency Numbers Module

Quick dial for Police, Ambulance, Fire, etc.

5. Nearby Help Module

Opens Google Maps showing nearby police stations and hospitals.

6. Articles Module

Displays women safety articles and real-life case studies.

7. Videos Module

Provides self-defense tutorial videos (via YouTube/WebView).

8. Review & Experience Sharing Module

Users can share their unsafe experiences (place, incident).

Other users can read experiences to stay alert.

5. Technology Stack

Frontend: Java (Android SDK), XML (UI Design)

Backend: Firebase (Authentication & Firestore Database)

APIs/Services:

Google Maps API (for location & nearby help)

Fused Location Provider (for live location tracking)

YouTube / WebView (for tutorial videos)

SMS Manager (for emergency alert messages)

6. Advantages

Provides real-time help in emergencies.

Easy-to-use interface for women of all age groups.

Increases awareness with articles and videos.

Community-driven review section helps identify unsafe areas.

Lightweight and efficient, works on most Android phones.

7. Applications

Personal safety for women, students, and working professionals.

Can be used by parents to ensure children's safety.

Useful in dangerous or isolated areas.

8. Conclusion

This Women Safety Application will act as a digital safety companion for women by combining emergency alert systems, location services, awareness resources, and community support. With its easy-to-use interface and real-time features, it contributes to creating a safer environment for women.